superscript:

$$x^{4+x}$$

$x = x^2$ subscript:

$$x_{2y} + 3$$

tragnomitric:

$$\sin x\pi$$

squarroot:

$$\sqrt[3]{x+2}$$

fractions:
the $\frac{2}{3}$ of the glass is full

$$\sqrt{\frac{1}{x^2 + \sin x}}$$